Exadata Database Services on OCI

Management Interfaces

Operations Advisory Team

Office of CTO | EMEA Solution Engineering

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Agenda

- Introduction
- Service Overview
- Management Interfaces
  - Control Plane based: UI, OCI-CLI, REST API, SDK, IaC Tools
  - Guest VM Command Line Interface (CLI): dbaascli, exacli

Introduction

Introduction

This workshop provides an overview of Oracle Exadata Database Services on Oracle Cloud Infrastructure (OCI), including the different deployment models, key architectural components, and the management interfaces available for administration and automation.

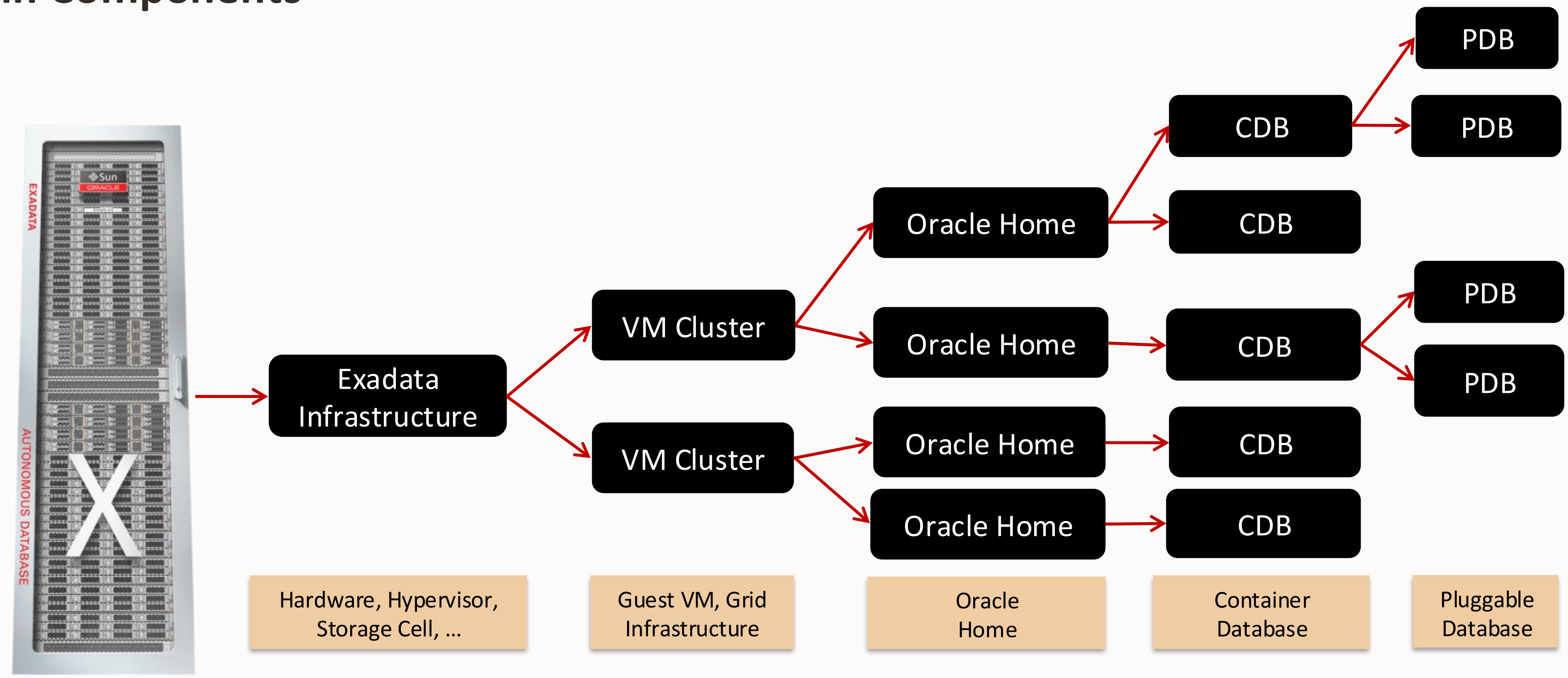
We will review the service architecture and explore the OCI and guest operating system interfaces used to manage Exadata resources, databases, and related services across the Exadata portfolio.

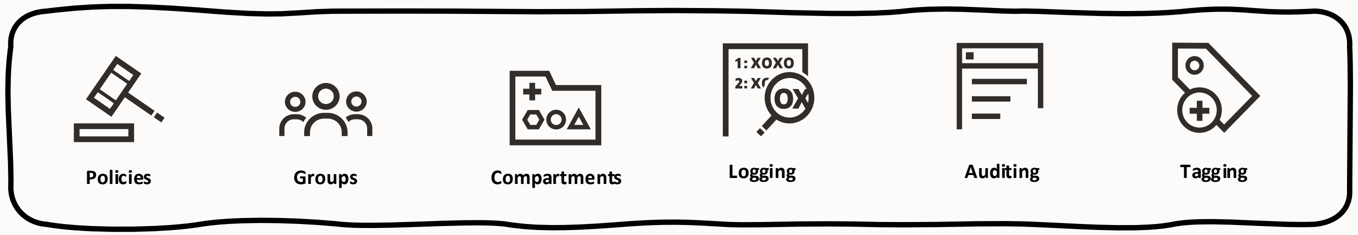
The code and commands presented in this workshop are intended as examples and may require adaptation for your specific environment.

Before executing any examples in your tenancy, review the official Oracle documentation and adjust all parameters, resource identifiers, and configuration values as appropriate for your deployment.

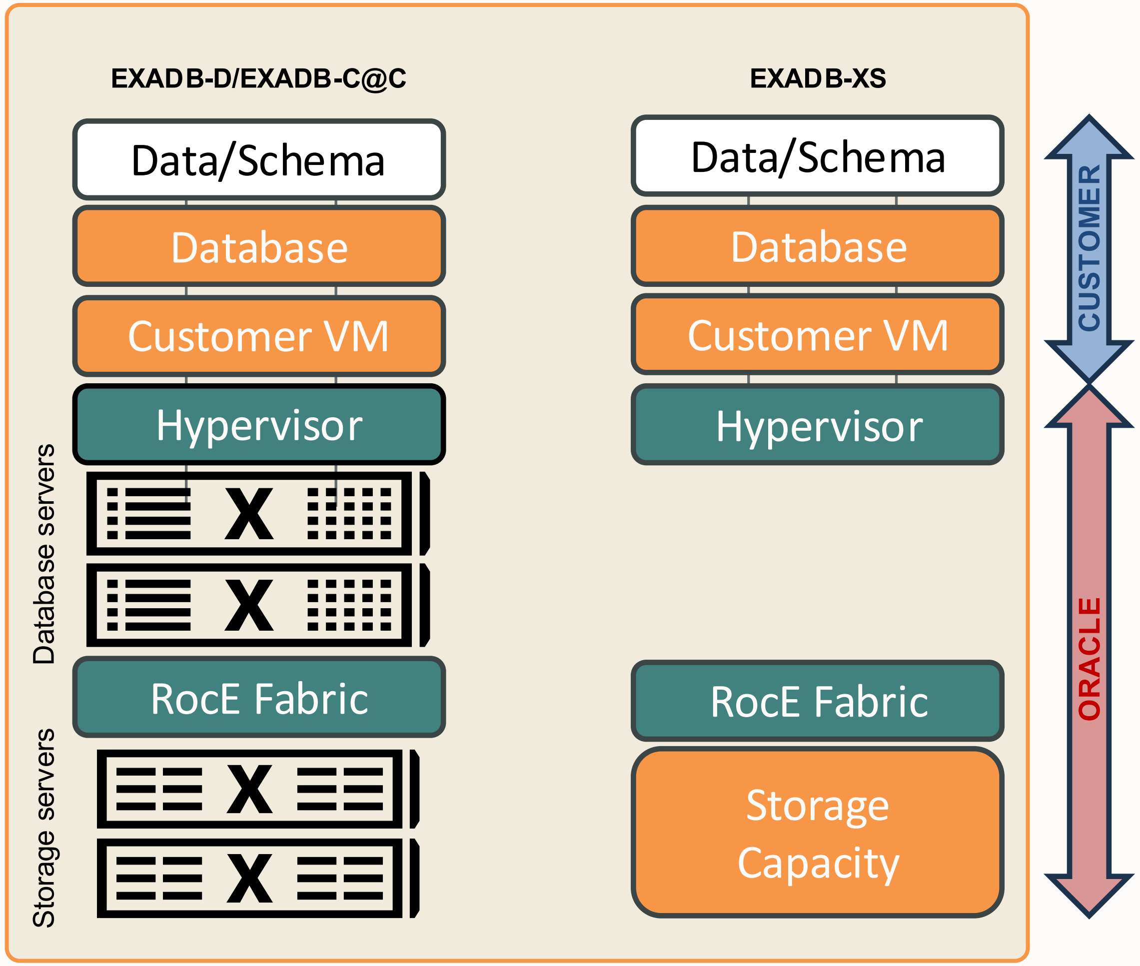
Service Overview

Main Components





Roles and Responsibilities



Function	Oracle Cloud Ops	Customer
Monitoring	Infrastructure, Control Plane, Availability, Capacity	Customer OS, Databases, Schemas
Incident Management & Resolution	Support for any incident related to the underlying platform	Incident Management and resolution for customer's Apps
Patch Management	Proactive patching of infrastructure	Patching of tenant instances and testing
Backup & Restoration	Provide running and customer accessible VMs.	Backup & Recovery of customer's data using Oracle native or 3rd party capability
Cloud Support	Response & Resolution of SRs	Submit SRs via MOS/Support Portal

Management Interfaces

Management Interfaces

The Exadata Cloud resources on Oracle Cloud Infrastructure (OCI) are created and managed through a variety of interfaces provided to fit your different management use cases:

OCI Control Plane Interfaces, eg UI, OCI-CLI, REST API, etc…

- Use OCI APIs to manage Exadata Cloud resources.
- Provide a consistent OCI IAM security model.
- Support interactive administration and automation.

Local VM Command-Line Interfaces, eg dbaascli, exacli, etc…

- Local to the guest VMs (no reliant on Control Plane connection)
- Support database and infrastructure administration operations.
- Complement OCI Control Plane interfaces.

Best practices on how these interfaces align for various Exadata Cloud database management use cases: Cloud API/CLI Alignment Matrix (Doc ID 2768569.1)

Management Interfaces

Control Plane Interfaces

OCI Control Plane Interfaces

OCI APIs

REST APIs that use HTTPS requests and responses for managing OCI resources, including Exadata Cloud

OCI Console

An intuitive, graphical interface for creating and managing your Exadata Cloud and other OCI resources, is one of the interfaces to the OCI APIs

Command Line Interface (OCI CLI)

Lightweight tool for command-line tasks. Supports scripts and extends Console functionality

Software Development Kit (SDK)

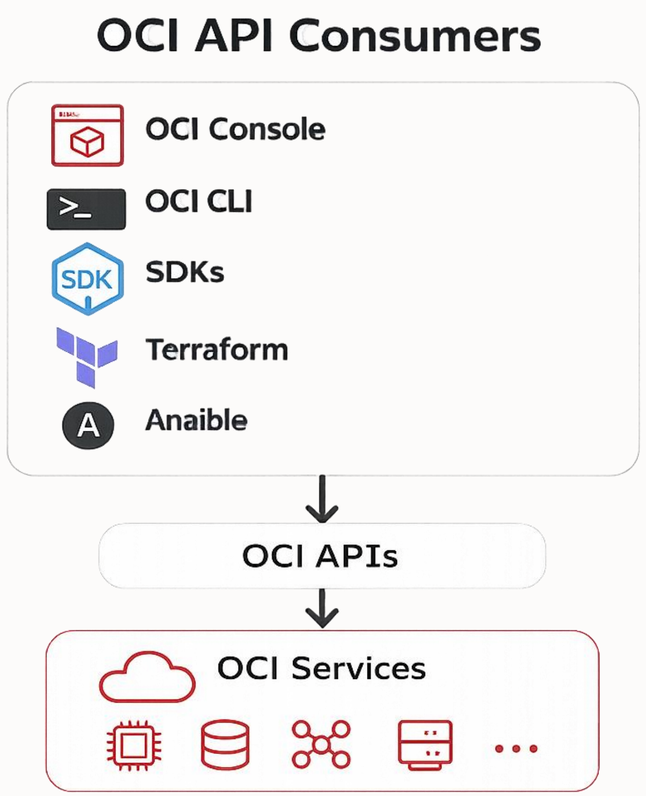
Enable custom solutions for Exadata Cloud and OCI services

IaC Tools & Plug-ins

Tools like Terraform Provider and Ansible Collection for provisioning and automation

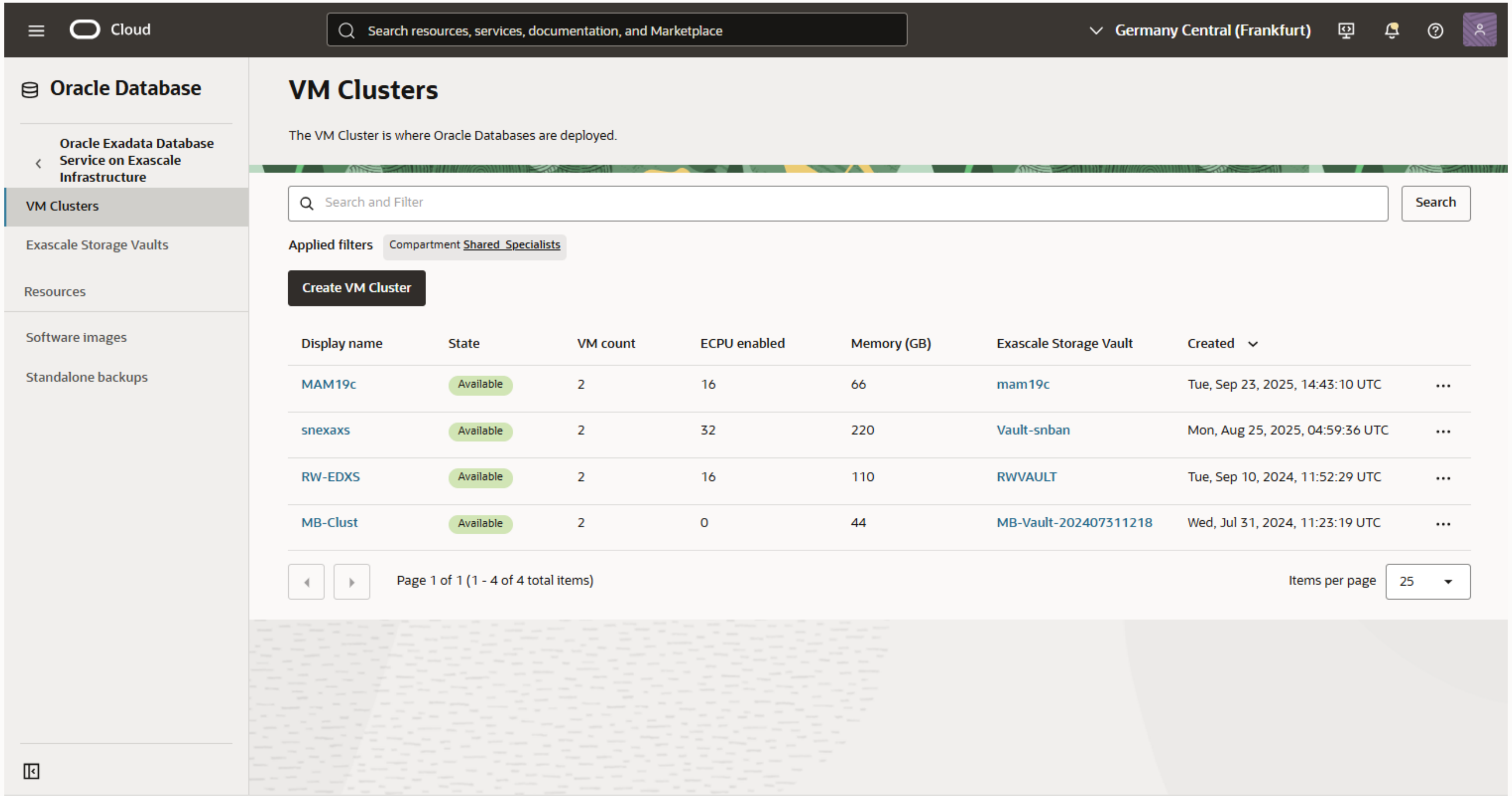
Cloud Shell

Browser-based terminal with pre-authenticated CLI, useful for tutorials and quick commands



OCI Console

Navigation Menu → Oracle Database → Oracle Exadata Database Service options



Command Line Interface (OCI CLI)

Sintax

```
$ oci [OPTIONS] COMMAND [ARGS] ...
Where COMMAND usually translates for:
<service> <type> <action> <options>
```

Help

```
$ oci --help
$ oci db -h
$ oci db system -h
```

Use environment variables

```
$ export T=<tenancy OCID>
$ export C=<compartment OCID>
$ oci iam compartment list -c $T
```

Authentication

```
$ oci session authenticate
$ oci iam region list --profile <profile> --auth security_token
```

A web browser window will be open asking for OC console login. Once we've entered our user details the windows could be closed

Input

```
$ oci [OPTIONS] COMMAND [ARGS] --from-json [text] ...
```

Inputs with more that one value, are passed in JSON format and can be provided as a string at the command line or as a file.

The ***--generate-full-command-json-input*** option can be used to generate a sample json file to be used with this command option

Output format

```
$ oci iam region list
or
$ oci iam region list --output table
```

By default, all responses are returned in JSON format. Depending on the output, it could be better to ask for a table format output with "--output table"

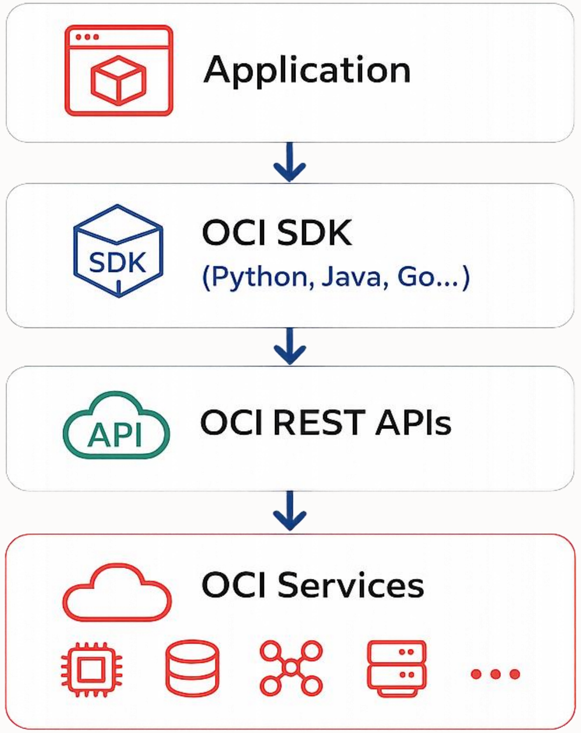
Software Development Kits (SDK)

Programmatic access to OCI services using your preferred programming language

Provide a programmatic way to manage Exadata Database Service on OCI. It supports multiple programming languages, making it flexible for automation and integration needs.

- Multi-Language Support:
  - Java, Python, Typescript/Javascript, .NET, Go, Ruby, PL/SQL
- Typical Use Cases:
  - Develop custom applications that interact with OCI
  - Integrate OCI operations into enterprise platforms
  - Build self-service portals and operational tooling
  - Develop new functionalities

OCI SDKs are wrappers around OCI REST APIs that simplify development and provide language-specific abstractions.



Resources:
- SDK Documentation: Oracle Cloud SDKs.
- Example Code: Available on GitHub for each language.

IaC Tools and Plug-ins

Infrastructure as Code (IaC) and Automation

Infrastructure as Code (IaC) enables the provisioning and management of OCI resources through machine-readable definitions and automation workflows.

Tool	Approach	Typical Use
Terraform	Declarative	Infrastructure provisioning
Ansible	Procedural	Operational Automation

Key Considerations

- Terraform is commonly used to provision OCI resources.
- OCI Resource Manager provides Terraform As-a-Service.
- Ansible complements Terraform for operational workflows.
- All approaches can be integrated into CI/CD pipelines.

IaC Tools and Plug-ins

Infrastructure as Code and Automation Tools

Terraform

Declarative Infrastructure as Code

- Define OCI resources using HCL
- Version-controlled infrastructure definitions
- Suitable for provisioning and infrastructure changes
- Supports OCI services through the OCI Provider

Typical Use Case:

- Provision and modify OCI infrastructure.

Ansible

Procedural Automation

- Automates operational workflows and runbooks
- Uses reusable playbooks
- OCI automation through the OCI Ansible Collection
- AWX / Ansible Automation Platform integration

Typical Use Case:

- Configuration management and operational automation.

Cloud Shell

What is Cloud Shell?

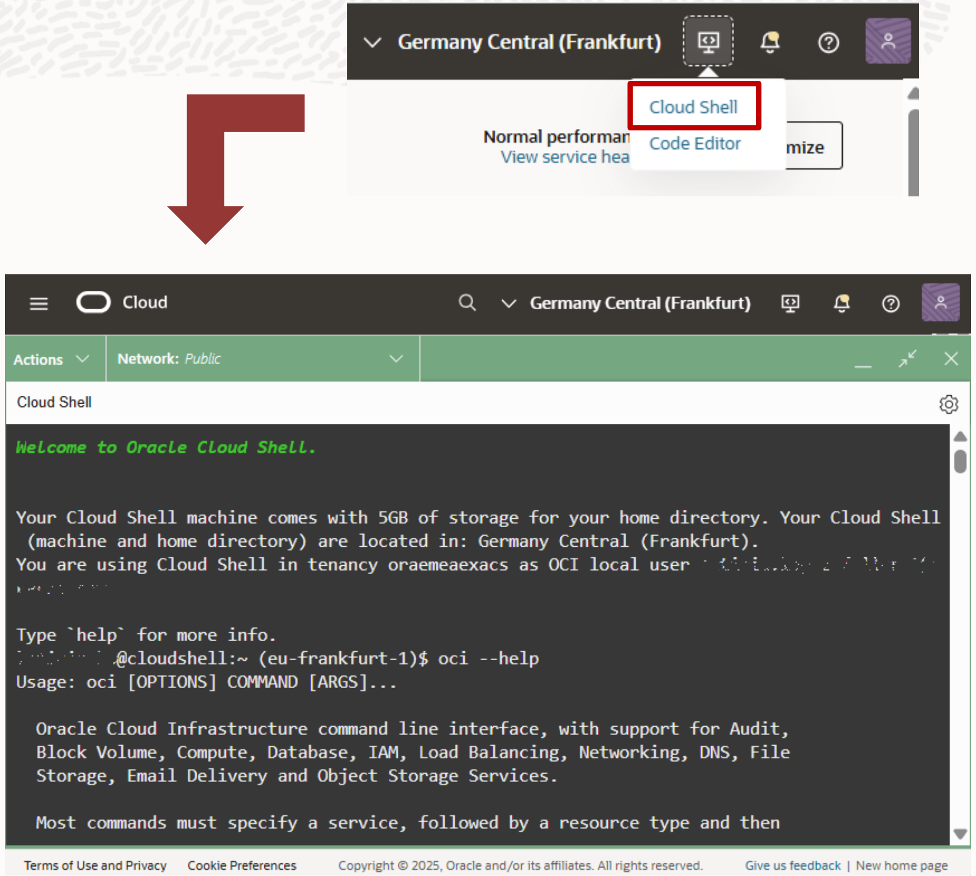
- A browser-based terminal in Oracle Cloud Console.
- Provides a pre-configured Linux shell.
- Includes Oracle Cloud Infrastructure (OCI) CLI, Ansible, and other tools.
- Free to use (within monthly tenancy limits).

Key Features

- Ephemeral VM for running commands.
- 5GB of encrypted persistent storage.
- Persistent frame stays active while navigating the Console.
- Accessible directly from the Console, no setup required.

Limitations

- No root access or sudo commands.
- Maximum session length: 24 hours (auto logoff after 60 minutes of inactivity).
- No external connections (e.g., SSH access not allowed).
- Cannot mount additional storage or use ping command.



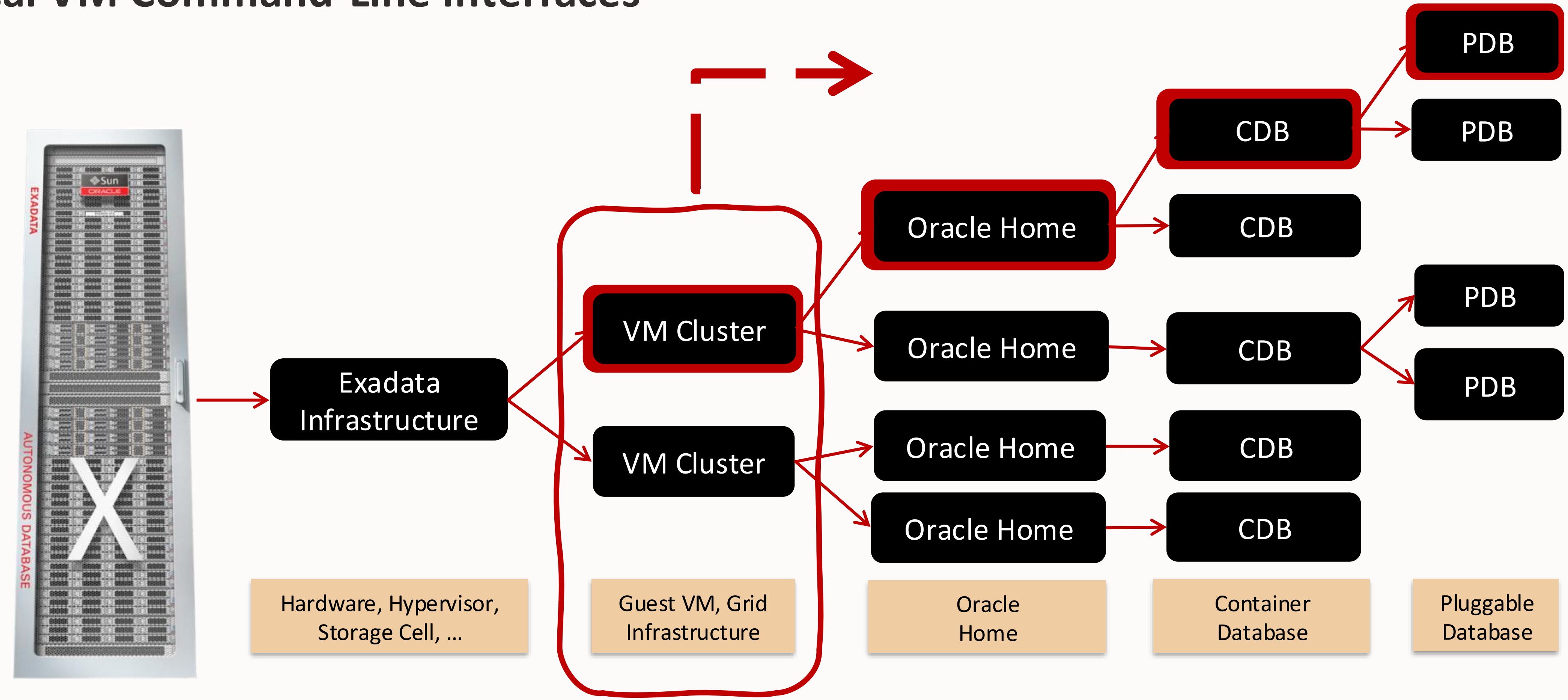
Pre-installed Tools:

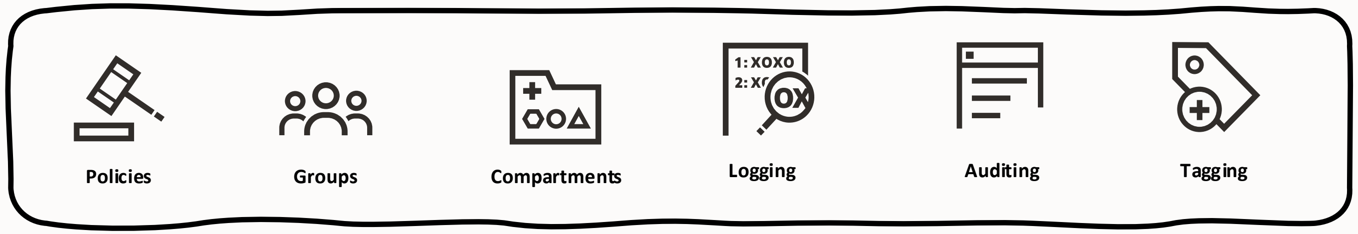
- OCI CLI, Git, Java, Python, Terraform, Ansible, Kubernetes tools (kubectl, helm), Docker, and more.
- Useful SDKs: Java, Python, Go, TypeScript, etc.

Management Interfaces

Local VM Command-Line Interfaces

Local VM Command-Line Interfaces





Local VM Command-Line Interfaces

CLI utilities located on the VM guests

Provisioned as part of the VM clusters on the Exadata Cloud Infrastructure, are available to perform various lifecycle and administration operations

dbaascli

Use the dbaascli utility to perform various database lifecycle and administration operations on the Oracle Exadata Database Service such as changing the password of a database user, starting a database, managing pluggable databases (PDBs), and more.

ExaCli

Use the ExaCLI command-line utility to perform monitoring and management functions on Exadata storage servers in the Exadata Cloud. (except ExaDB-XS)

bkup_api

DEPRECATED

Handles backup/recovery tasks, including on-demand database and PDB backups.

Must be connected to a VM in the Exadata Cloud VM cluster; uses VM OS user security, not OCI user security.

Changes sync to OCI Control Plane via ***DB Sync***.
Tools are auto-updated by Oracle on a regular basis.

dbaascli operations

- Administration and Configuration
- Cloud Software Library (CSWLIB) Management
- Database Management
- Data Guard Management
- Database Home Management
- Diagnostics and Health Checks
- Grid Infrastructure Management
- Patching and Upgrading
- Pluggable Database (PDB) Management
- System Management
- Transparent Data Encryption (TDE) Management
- …

dbaascli examples

List all commands

```
# dbaascli list [-help]
```

Show latest dbaastool available

```
# dbaascli admin showLatestStackVersion
```

Install/update a dbaastools RPM

```
# dbaascli admin updateStack
```

Display details about a specific database

```
# dbaascli database getDetails --dbname <db_name>
```

Get the list of PDBs in the database

```
# dbaascli database getPDBs --dbname <db_name>
```

Checks status of a database

```
# dbaascli database status --dbname <db_name>
```

Display Oracle Homes

```
# dbaascli system getDBHomes
```

Display Oracle Homes Details

```
# dbaascli dbHome getDetails --oracleHomeName <db_home>
```

List the details of all the Grid homes

```
# dbaascli system getGridHomes
```

List available DB/GI images

```
dbaascli cswlib showImages --product [GRID|DATABASE]
```

Create a database

```
# dbaascli dabase create -dbName <db_name> 1…
```

Database out-of-place patching

```
# dbaascli database move -dbname <db_name> 1…
```

[1]… Not all parameters listed. Please check the command definition

Native Oracle Administration Tools

Native Oracle administration tools remain available but should only be used when the required operation is not supported through cloud management interfaces, or when explicitly documented by Oracle.

Tool	Typical Use
SQL*Plus / SQLcl	Database administration and SQL execution
RMAN	Backup and recovery operations
DGMGRL	Advanced Data Guard administration
ASMCA / SRVCTL	Grid Infrastructure administration
Data Pump	Data migration and export/import

Use cloud management interfaces whenever the required operation is supported.

- Maintains OCI automation and lifecycle management
- Ensures consistency between the guest VM and OCI Control Plane
- Reduces operational risk
- Preserves supportability
- Manual changes performed outside supported cloud management interfaces **may not be reflected** in the OCI Control Plane.

Summary

Tool	Category	Typical Use
OCI Console	OCI Control Plane	Interactive Administration
OCI CLI	OCI Control Plane	Automation & Scripting
REST APIs	OCI Control Plane	Integration & Custom Tooling
SDKs	OCI Control Plane	Application Development
Cloud Shell	OCI Control Plane	Browser-Based Operations
Terraform	OCI Control Plane	Infrastructure Provisioning
OCI Ansible Collections	OCI Control Plane	Operational Automation
dbaascli	Guest VM Interface	Database Lifecycle Management
EXACLI*	Guest VM Interface	Exadata Infrastructure Administration

* Except ExaDB-XS

Stay tuned

[Operations Advisory GitHub Repository]

Thank you

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